

Strategic Visualization Storyboard

Hierarchical Data Analysis for Global Expansion

Decision Support for Terra Cotta Foods

End User: Marco Antonelli
Role: Chief Executive Officer (CEO)
Company: Terra Cotta Foods (TCF)

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Executive Summary

This storyboard presents a hierarchical data visualization strategy designed specifically for Marco Antonelli, CEO of Terra Cotta Foods (TCF), to support strategic decisions regarding global expansion into Latin America and Asia.

Strategic Context

Terra Cotta Foods is planning an ambitious expansion strategy to diversify suppliers, secure supply chains, and mitigate geopolitical risks. The visualization must enable rapid validation of business hypotheses and clear identification of investment priorities.

1 User Persona: Marco Antonelli

Profile

Name: Marco Antonelli

Position: Chief Executive Officer (CEO)

Company: Terra Cotta Foods (TCF)

Industry: Food and Beverage Distribution

Company Type: Major distributor of premium food products and raw materials

Background

Marco Antonelli is the CEO and driving force behind Terra Cotta Foods (TCF), one of Europe's largest distributors of premium food products and raw materials. TCF's core business revolves around acquisition, processing, and distribution of goods, making logistical efficiency and cost control vital to operations.

Marco is currently planning an ambitious expansion strategy into Latin America and Asia to diversify suppliers, secure supply chains, and mitigate geopolitical risks.

Data Literacy Level

High at operational and financial levels, but requires complex data to be distilled into actionable business metrics. Marco values tools that save time and enable rapid validation of business hypotheses.

Key Characteristics

- **Strategic Thinker:** Focuses on long-term competitive positioning
- **Results-Oriented:** Demands clear, actionable insights
- **Time-Constrained:** Needs rapid decision-making tools
- **Risk-Aware:** Considers geopolitical and supply chain vulnerabilities

2 Business Challenge

Strategic Question: Which regions and countries offer the best opportunities for TCF's expansion in terms of purchasing power, market size, and supply chain viability?

Primary Objectives

- 1. Market Assessment (Countries/GDP/Population):**
Determine the economic viability and potential reach of TCF's supply chain in new markets.
- 2. Purchasing Power Hierarchy:**
Which regions and countries show the highest GDP (purchasing power/buying capacity) to justify investment in distribution centers?
- 3. Supply Risk Analysis:**
How does population (labor force/consumer base) correlate with economic potential in countries where TCF plans to establish sourcing facilities or factories?
- 4. Internal Resource Allocation:**
Which business lines need to be adjusted internally to support the expansion strategy?

3 Visualization Requirements

Marco's Expectations

The visualization must act as a strategic roadmap. The hierarchical data presentation should be clear enough to enable decisions like "Do we invest in Continent/Country X or Y?" and "Which business line needs internal adjustment?" The visualization must prioritize clarity, information hierarchy, and robust design that conveys confidence and control over the complex global supply chain.

Critical Success Factors

- **Clarity:** Immediate understanding without extensive training
- **Hierarchy:** Clear continent-to-country drill-down capability
- **Actionability:** Direct path from insight to decision
- **Confidence:** Professional design that supports boardroom presentations

4 Data Analysis Insights

Based on the comparative analysis conducted across three datasets, the following insights are directly relevant to Marco's decision-making needs:

4.1 Global Population Distribution (Treemap Analysis)

Continent	Population Share	Market Size	Strategic Value
Asia	58.9%	Massive	High
Africa	18.1%	Growing	Emerging
Europe	9.4%	Mature	Established
North America	7.6%	High GDP	Premium
South America	5.5%	Moderate	Target
Oceania	0.6%	Small	Limited

Table 1: Population distribution and strategic assessment by continent

Key Insight for Marco:

Asia dominates with 58.9% of global population, presenting the largest consumer base. However, North America leads in GDP per capita despite smaller population, indicating different strategic approaches are needed for each region.

4.2 Economic Clustering (Dendrogram Analysis)

Countries were grouped into five strategic clusters based on population and GDP per capita:

- Population Giants:** China, India
Strategic Value: Massive consumer base, competitive labor costs, high sourcing potential
- High GDP Micro-states:** Monaco, Liechtenstein
Strategic Value: Premium markets, limited supply chain value
- Developing Economies:** Brazil, Russia, Mexico, Indonesia
Strategic Value: Balanced market size and growth potential, ideal for expansion
- Advanced Economies:** USA, Switzerland, Singapore
Strategic Value: High purchasing power, established distribution infrastructure
- Developed Nations:** Germany, Japan, UK, France
Strategic Value: Mature markets, efficient supply chains, premium segments

Recommendation for TCF

Focus expansion efforts on Cluster 3 (Developing Economies) for optimal balance between market potential and investment requirements. Use Cluster 1 (Population Giants) for sourcing and manufacturing, while maintaining premium distribution channels in Clusters 4 and 5.

4.3 Resource Allocation Model

Drawing from the organizational structure and budget distribution analysis:

Business Unit	Current	Recommended	Justification
Supply Chain	36%	40%	Expansion priority
Market Development	24%	28%	New territory entry
Operations	24%	20%	Efficiency gains
Product Innovation	16%	12%	Mature portfolio

Table 2: Recommended internal resource reallocation for expansion

5 Visualization Strategy

5.1 Recommended Approach: Multi-Layer Treemap

Based on the analysis, a **Treemap** is the optimal visualization for Marco’s needs:

- **Size Encoding:** Rectangle area represents GDP (purchasing power)
- **Color Encoding:** Heat map indicates GDP per capita (market maturity)
- **Hierarchy:** Continent → Country → Economic Cluster
- **Interactivity:** Click-through for detailed metrics

5.2 Why Treemap Over Other Methods?

Method	Advantages	Limitations for TCF
Treemap	Proportional, compact, clear hierarchy	None significant
Dendrogram	Shows clustering well	Lacks quantitative scale
Sunburst	Radial hierarchy	Label overlap issues
Bar Chart	Simple comparison	No hierarchical context

Table 3: Visualization method comparison for Marco’s use case

6 Storyboard: The Decision Journey

Scene 1: The Strategic Question

Marco opens the dashboard before a board meeting...

Marco’s Thought: “We need to decide between Latin America and Asia for our next major expansion. Where do we get the best return on investment?”

Scene 2: Continental Overview

Visual: Treemap showing all continents

Insight: Asia dominates in size (58.9% population), but North America shows deepest green (highest GDP per capita)

Action: Marco clicks on Asia and South America to compare

Scene 3: Regional Drill-Down

Visual: Side-by-side comparison of Asian and South American countries

Insight:

- China and India offer massive consumer bases
- Brazil and Mexico show balanced market size with growth potential
- Argentina and Colombia represent emerging opportunities

Action: Marco identifies target countries in each region

Scene 4: Economic Cluster Analysis

Visual: Countries color-coded by economic cluster

Insight: Developing economies (Cluster 3) offer optimal risk-reward balance

Decision Point: "We'll establish distribution centers in Brazil and Indonesia, with sourcing facilities in India"

Scene 5: Resource Allocation

Visual: Internal budget treemap showing current vs. recommended allocation

Insight: Supply chain needs 4% increase, funded by efficiency gains in operations

Action: Marco approves the strategic plan for board presentation

7 Design Specifications

7.1 Visual Design Requirements

Color Scheme

- **GDP Scale:** Green gradient (light = low, dark = high)
- **Regions:** Distinct colors per continent
- **Highlights:** Orange for target markets
- **Alerts:** Red for high-risk areas

Typography

- **Headers:** Bold, sans-serif
- **Data Labels:** Clear, readable at 10pt
- **Tooltips:** Detailed metrics on hover
- **Legends:** Intuitive placement

7.2 Interaction Design

- 1. **Initial View:** Continental treemap with GDP and population
- 2. **Click Action:** Zoom into country-level detail
- 3. **Hover State:** Display key metrics (GDP, population, GDP per capita)
- 4. **Comparison Mode:** Side-by-side region analysis
- 5. **Export Function:** PDF report for board meetings

8 Key Performance Indicators

The visualization must surface these metrics instantly:

Metric	Business Question	Target
GDP (Total)	Market size	≥\$1T
GDP per Capita	Purchasing power	≥\$10K
Population	Consumer/labor base	≥50M
Economic Cluster	Market maturity	3-4
Supply Chain Risk	Geopolitical stability	Low-Medium

Table 4: Decision metrics for market selection

9 Success Criteria

Visualization Success Metrics

- 1. **Speed:** Marco can identify top 3 target countries in under 2 minutes
- 2. **Confidence:** 95% certainty in decision based on visual insights
- 3. **Clarity:** Zero questions about data interpretation in board meeting
- 4. **Actionability:** Direct path from visualization to strategic plan

10 Implementation Roadmap

- 1. **Phase 1:** Static treemap with continental overview (Week 1)
- 2. **Phase 2:** Interactive drill-down capability (Week 2)
- 3. **Phase 3:** Comparison mode and filtering (Week 3)
- 4. **Phase 4:** Export and presentation features (Week 4)
- 5. **Phase 5:** User testing with Marco and refinement (Week 5)

11 Conclusions

Strategic Alignment

The hierarchical visualization approach directly addresses Marco Antonelli’s need for rapid, confident decision-making in TCF’s global expansion strategy. By leveraging insights from our comparative analysis:

- Asia’s 58.9% population share identifies sourcing opportunities
- North America’s GDP leadership confirms premium market strategy
- Developing economy cluster highlights expansion targets
- Resource allocation model guides internal adjustments

Expected Business Impact

Strategic Benefits

- Faster market selection
- Reduced decision risk
- Clearer investment priorities
- Better board communication

Operational Benefits

- Optimized resource allocation
- Aligned expansion timeline
- Quantified ROI projections
- Trackable success metrics

Final Recommendation

Implement a multi-layer interactive treemap as the primary decision support tool for TCF’s global expansion strategy, complemented by cluster analysis overlays and comparative views. This approach combines the clarity Marco needs with the analytical depth required for confident boardroom decisions.

The Bottom Line

The right visualization doesn’t just present data—it enables decisions. For Marco Antonelli and Terra Cotta Foods, this hierarchical approach transforms complex global market data into a clear strategic roadmap.

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CountriesClustersTarget RegionsStrategic Vision

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Phase 1: Portfolio Checkpoint 2